

AvatarX — 20-Milestone Plan: Proof of Concept → Clinical Grade

Audience: Engineering team · **Date:** August 15, 2026 · **Status:** v1.0 for review **Repo:** AvatarX/code/afib (extend in place) · **Scope:** five biomarker tracks — AFib, ECG (inferred/gated), arterial stiffness, vascular tone, VO₂max — on one shared platform. **Companion docs (repo):** feasibility review + brief (AvatarX/research/), implementation plan (docs/), execution prompts v0.2–v0.5.

0. How to read this plan

The maturity ladder every biomarker climbs (a milestone = moving one track up one rung, on evidence):

POC (composes on synthetic/public data) → Prototype (runs on real paired first-party data, honest metrics, gates fire) → Validated (beats all baselines on unseen subjects, subject+session-disjoint) → Clinical-Grade (locked model, prospective + external validation, regulatory pathway engaged).

One rule governs the whole ladder: a track advances only when it beats its mandatory baselines on data it has never seen, evaluated by the production path — never on a side harness, never on enriched cohorts. Milestones are dependency-ordered, not calendar-dated; the long poles (clinical recruitment, regulatory, blinded reads) are flagged. Sizing is relative (S/M/L/XL).

Honest per-track ceilings (engineering must design to these, not around them):

Track	Realistic clinical-grade endpoint	Physics ceiling
AFib / rhythm	Contactless AFib screening (confirm w/ ECG)	Strong — rides beat timing, which survives to the face
ECG (inferred)	Rhythm conclusions; waveform stays gated research	Morphology (P/QT/ST) physically absent from facial video — reconstruction gates predicted to stay red
Arterial stiffness	Stiffness index / wellness "vascular age" trend	Morphology-fragile through camera; must beat age+BP baseline
Vascular tone	Wellness reactivity trend vs own baseline	Amplitude is optics-confounded; validate by provocation, not absolute level
VO ₂ max	Fitness estimate (wellness); submax-response version later	Resting scan largely proxies age/HR/demographics — the weakest of the five; needs an exercise/HR-recovery signal for real physiology

PHASE A — Prototype the five tracks on real data (Milestones 1–6)

M1 — Biomarker POC→Prototype: unify all five heads on the platform. Land AFib, ECG-reconstruction (gated), stiffness, tone, and VO₂max as registered heads/ plug-ins on the shared BeatLattice, each producing a HeadResult with measurement class and confidence, each with a synthetic test fixture so the full pipeline + gates run in CI. *Exit:* cli.py process --heads afib,ecg,stiffness,vasotone,vo2 runs end-to-end on synthetic + public fixtures; all existing tests green; every head's forbidden-output/quarantine tests pass. *Size:* L. *This is milestone one as requested.*

M2 — Signal-fidelity gate for every morphology track (the go/no-go science). Run the v0.4/v0.5 fidelity harness: which features (stiffness morphology, tone amplitude/notch, VO₂max HR-recovery proxies) actually survive our camera pipeline vs simultaneous contact reference, per Fitzpatrick group and device. *Exit:* per-feature ICC vs contact + test-retest scoreboard published; features below threshold auto-dropped by config. *Size:* M. *Dependency for M9–M11; a track with no surviving features stops here — an honest, cheap "no."*

M3 — Capture hardening for clinical acquisition. 60 fps target, per-frame hardware timestamps, AE/AWB lock where OS permits, pre-encode ROI processing, PRBS/LED sync verified <10 ms in field conditions; consumer + reference capture profiles. *Exit:* sync error measured <10 ms on-rig; device matrix (≥3 phone families) characterized. *Size:* M.

M4 — Reference-ground-truth ingestion for all five. cli.py ingest-reference accepts: continuous single-lead ECG (AFib/ECG), cfPWV/CAVI export (stiffness), pulse-oximeter perfusion-index trace (tone), and CPET/estimated-VO₂max + HR-recovery (VO₂max) — each with device/operator metadata, sync verification, and episode/phase-locked labels. *Exit:* one multi-reference session ingests cleanly with all labels; fail-closed on sync error. *Size:* L.

M5 — Data engine: registry, campaigns, disjoint splits. `datasets/registry.jsonl` (content hashes, consent class, lineage); campaign manifests with Fitzpatrick/device/age/BP quotas; participant + session-disjoint (+ leave-one-device/site-out) splits. *Exit:* `cli.py campaign status` reports quota fill; splits provably disjoint (hash test). *Size:* M.

M6 — Training engine + model registry + promotion gate. Config-driven deterministic runs; `models/registry.jsonl` (semver, lineage, eval-report hash); `cli.py promote` refuses on any red gate. *Exit:* a retrain→evaluate→promote cycle runs end-to-end on fixtures; promotion writes an auditable record. *Size:* M.

PHASE B — First real cohort & per-track baseline defenses (Milestones 7–12)

M7 — Pilot capture cohort (feasibility $N \approx 60\text{--}100$, multi-tone, multi-device). The first first-party paired dataset; recruit for diversity, not convenience. *Exit:* beat-timing IBI MAE ≤ 30 ms at rest across Fitzpatrick I–VI; type-VI coverage $\geq 70\%$; no-read rates characterized. *Size:* L (recruitment-gated — long pole).

M8 — AFib head → Validated on the pilot + public teachers. Retrain on degraded-real + first-party RR; must beat tachogram-statistics and demographic baselines subject-independent. *Exit:* AUC ≥ 0.90 on unseen subjects with significant baseline deltas; leakage audit clean. *Size:* M.

M9 — Arterial-stiffness head → beats the age+BP baseline (T2 defense). Model on V0-surviving features vs cfPWV; must beat age+sex+brachial-BP regression, not just age. *Exit:* incremental-validity margin met on held-out subjects, or track paused with documented negative result. *Size:* M.

M10 — Vascular-tone head → passes the optics null-arm + HR/resp baselines. Provocation protocol (cold pressor / paced breathing / null_optics / null_rest); tone response must track contact-PI and stay flat in the lamp-only arm and beat HR/respiration-response baselines. *Exit:* §W W0–W2 green on pilot, or paused with negative result. *Size:* M.

M11 — VO₂max head → beats the demographics+resting-HR baseline (hardest bar). The honest test: does the *scan* add fitness information beyond age/sex/BMI/resting-HR? If not, it is a demographic calculator and must be labeled/paused as such. Add an optional submaximal HR-recovery capture mode (short guided step/recovery) as the physiologically-defensible path. *Exit:* scan beats the demographic+HR baseline on unseen subjects, OR VO₂max is re-scoped to the HR-recovery-mode research track. *Size:* M–L.

M12 — ECG-reconstruction falsification lab reports (gated track truth). Train the decoder on first-party paired data; run identity-baseline, interval-error, morphology, and confabulation gates (§G). *Exit:* gate-status scoreboard on real data published; expected outcome — morphology gates red — documented as AvatarX's evidence asset; rhythm detection stays on the measured path. *Size:* M.

PHASE C — Robustness, fairness, and quality at scale (Milestones 13–16)

M13 — Signal-quality & no-read architecture productionized. Three-state gate (ACCEPT / REPEAT_SCAN / NO_RESULT) with per-scan and per-beat confidence, cross-ROI and rBCG-concordance checks (rBCG ablation-only), fairness-aware coverage. *Exit:* accuracy-vs-coverage curves per track; per-Fitzpatrick coverage floor enforced as a release gate. *Size:* M.

M14 — Cross-device / real-world robustness campaign. Multi-phone, multi-lighting, ambulatory, home conditions; leave-one-device/site-out evaluation. *Exit:* each Validated track holds within pre-registered margins across strata; degradations quantified, not hidden. *Size:* L.

M15 — Bias & fairness validation (release-gating). Formal per-Fitzpatrick, age, sex, BMI error and coverage parity; darker-skin performance is a go/no-go, not a footnote (field no-read rates reach 39% — we must beat that). *Exit:* parity within bounds for every shipping track, or track ships with documented, disclosed limits. *Size:* M.

M16 — Hard-negative & confounder campaign (AFib-critical). Enrich for PAC/PVC, atrial flutter, sinus arrhythmia, motion, plus vasoactive-medication and BP-range coverage for the vascular tracks. *Exit:* AFib confusion matrix (AF vs each imposter) on unseen subjects; flutter/ectopy limits quantified and labeled. *Size:* L (clinical partnerships).

PHASE D — Clinical grade & regulatory (Milestones 17–20)

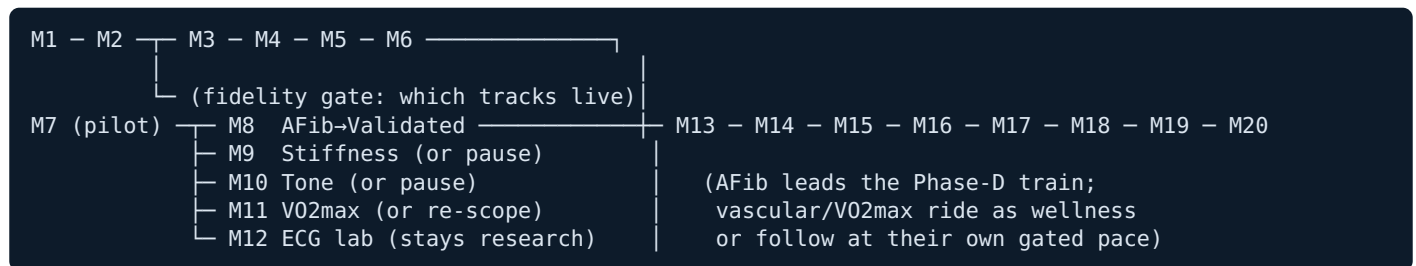
M17 — Lock models; pre-register the pivotal protocols. Freeze algorithms per track; pre-register endpoints, thresholds, and analysis plans (SAP) with the clinical advisor; no post-hoc tuning after enrollment. *Exit:* locked model hashes in registry; protocols registered. *Size:* M.

M18 — Prospective pivotal validation (per shipping track). AFib first (adjudicated ECG comparator, deployment-prevalence PPV, confirmatory-ECG funnel); vascular/VO₂max tracks at their gated wellness or screening claim. *Exit:* pre-registered Se/Sp/PPV/NPV (or reference-agreement) met on prospectively enrolled, unseen subjects. *Size:* XL (clinical — longest pole).

M19 — Independent external validation. New site(s), investigators, devices, population; model unchanged. *Exit:* replication within CI of pivotal; results published whichever way they fall. *Size:* L.

M20 — Regulatory submission & QMS. FDA Q-Sub already engaged (start at M6–M8); assemble the submission for the lead claim (contactless AFib screening — De Novo likely, no predicate exists), with a Predetermined Change Control Plan (PCCP) for model updates and a design-history/QMS trail from the registries built in M5–M6. *Exit:* submission filed for the lead claim; wellness tracks (tone/VO₂max trends) shipped under General Wellness in parallel. *Size:* XL (regulatory).

Dependency spine & parallelism



Phase A (M1–M6) has no clinical dependency — start immediately. Phases B–D are paced by recruitment (M7, M16, M18) and regulatory (M20), the two true long poles. Tracks are allowed to **fork**: AFib is expected to reach clinical grade first and pull the regulatory train; stiffness/tone/VO₂max advance as wellness features and only enter Phase D if their baselines are beaten; ECG-reconstruction is not expected to leave the falsification lab — and that is a planned, documented outcome, not a failure.

Team & governance

Signal/CV eng (L0–L3, capture) · ML eng (heads, training, reconstruction lab) · data/backend eng (registries, pipelines, splits) · capture/mobile-web eng (campaign tooling, demo) · clinical data coordinator (M7/M16/M18 campaigns, adjudication) · fractional clinical advisor (gate thresholds, blinded reads, SAP) · regulatory consultant (M6 Q-Sub → M20). Tests-first is the repo's law; every milestone's exit criteria are encoded as tests or a signed gate record before the milestone is called done.

Top program risks

Recruitment slower than plan (M7/M16/M18) — parallel site partnerships, passive-capture option. · Darker-skin performance gap — fairness as release gate (M15), not late QA. · Gate erosion under demo/commercial pressure — thresholds changeable only via signed spec changelog; forbidden-output tests in CI. · A track's fidelity gate fails (M2/M9–M11) — treated as a cheap, honest stop, re-scoped to wellness or shelved, freeing effort for AFib. · Regulatory path assumption wrong — De Novo vs 510(k) resolved early via Q-Sub, before the pivotal is designed.

Milestones are evidence-gated, not calendar-committed; figures cited elsewhere are controlled-study ceilings, not deployment guarantees. Numeric gate thresholds require clinical sign-off. This plan is engineering guidance, not a clinical or regulatory claim.